

Pool Game: Concurrent Implementation Report

Assignment 01 - PCD Course

Academic Year 2025-2026

Abstract

This report presents a concurrent implementation of the Pool Game, a 2D board game where two players (human and bot) control balls to push smaller balls into holes. The document provides a detailed analysis of concurrency challenges, design architecture, behavioral specifications using Petri Nets, performance evaluation, and formal verification using JPF (Java Path Finder). The concurrent implementation demonstrates significant improvements over a sequential baseline while effectively exploiting available processor cores through multithreading and task-based approaches.

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1 Problem Analysis

1.1 Problem Description

The Pool Game is a two-player billiard-like game on a 2D board with the following characteristics:

- **Dynamic Objects:** Multiple balls (typically ranging from 2 to 4500) moving and interacting on the board
- **Physics Simulation:** Elastic collisions, friction forces, and momentum transfer between bodies
- **Players:** Two players (human controlled via keyboard, bot controlled autonomously) with scoring systems
- **Game Objective:** Insert small balls into holes at board corners; winner is determined by score or by eliminating opponent's ball

1.2 Concurrency-Related Challenges

1.2.1 Computational Complexity

- **Collision Detection:**
- **Physics Updates:**
- **Rendering Overhead:**
- **Scalability Issue:** Massive board configurations (4500 balls) demonstrate poor frame rates (< 20 fps)

1.2.2 Temporal Constraints

- Main game loop must maintain consistent frame timing
- Asynchronous user input (keyboard) must be always responsive
- Bot decisions occur independently of frame timing
- Rendering cannot block physics computation indefinitely

1.2.3 Synchronization Requirements

- Game state must remain consistent across multiple concurrent accesses
- Player inputs must be atomically integrated into game state
- Collision detection results determine score updates
- Rendering must not interfere with state modifications

1.2.4 Data Races and Race Conditions

Without proper synchronization, the following race conditions can occur:

- **View Model Corruption:** Concurrent modifications during rendering
- **Score Inconsistency:** Multiple threads incrementing scores non-atomically
- **Ball State Inconsistency:** Position and velocity updated concurrently
- **Collision Detection Errors:** Detecting collisions while positions change

1.3 Task-based Related Challenges

The task-based concurrent implementation using the Executor Framework introduces a different set of challenges compared to the thread-per-entity model. The Pool implementation uses work-stealing task decomposition via `WorkerPool` to parallelize physics updates and collision detection.

1.3.1 Computational Complexity

Task-based parallelism in the Pool implementation must address several complexity-related challenges:

- **Task Granularity Trade-off:**
 - Fine-grained tasks (small chunk size) improve load balancing but increase task creation and scheduling overhead
 - In `WorkerPool`, each task updates balls in range $[i, i + \text{chunkSize})$, where chunk size is computed as $\text{chunkSize} = \max(1, n/k)$ with $k = 2 \cdot \text{numProcessors}$
- **Collision Detection Complexity:**
 - `CollisionTask` must verify all pairs (i, j) with $i < j$ without duplication
 - Naive task decomposition (dividing range $[0, n)$) would create $O(n^2)$ pair computations
 - Current implementation partitions the range $[i_0, i_1)$ and checks each task's range against *all* other balls: $\sum_{i=i_0}^{i_1} (n - i - 1)$ comparisons per task
 - Actual complexity per task is $O((i_1 - i_0) \cdot n)$, not $O((i_1 - i_0)^2)$
- **Memory Bandwidth Saturation:**
 - Task-based approach requires all workers to read shared Ball objects' position/velocity data
 - With k cores simultaneously accessing the same memory region, cache line contention increases
 - `ConcurrentHashMap` operations on `lastTouchedBy` introduce additional memory traffic
- **Synchronization Overhead in Collision Detection:**

```
1    synchronized (first) {  
2        synchronized (second) {  
3            // Atomic collision resolution  
4            Ball.resolveCollision(a, b);  
5        }  
6    }
```

Listing 1: Ordered Lock Acquisition

- Each collision check acquires locks on two Ball objects using `synchronized(first)` then `synchronized(second)`
- Lock ordering (by `System.identityHashCode()`) prevents deadlock but adds comparison overhead
- If two tasks try to resolve the same collision pair with different order, one will block the other

1.3.2 Temporal Constraints

Task-based parallelism introduces timing-related constraints that differ from the thread-per-entity model:

- **Barrier Synchronization Latency:**

- In `WorkerPool`, `PlayerCollisionTask`, `parallelBallUpdate()` and `parallelCollisionDe` use `invokeAll()`, which is a synchronization barrier
- All tasks must complete before control returns to the `GameLoop`
- Frame time = $\max(\text{task duration}) + \text{overhead}$

- **Task Queue Delay:**

- Created tasks are submitted to the `ExecutorService` queue
- If all workers are busy, new tasks wait in queue, delaying their execution
- With a fixed thread pool of size k , tasks may experience queueing delay proportional to queue depth
- This adds non-deterministic latency to each game frame

- **Future.get() Blocking Behavior:**

- The main `GameLoop` thread calls `f.get()` (code in class `WorkerPool`) on all futures from `invokeAll()`
- This is a blocking operation—the `GameLoop` cannot proceed until *all* tasks complete
- Any uncaught exception in a task will be re-thrown at `get()`, potentially crashing the loop

1.3.3 Synchronization Requirements

Task-based parallelism requires careful synchronization to ensure memory consistency and avoid data races:

- **ConcurrentHashMap for lastTouchedBy:**
 - Multiple `CollisionTask` instances concurrently update `lastTouchedBy` map via `put()` calls
 - Using a non-concurrent map would cause data races
 - `ConcurrentHashMap` uses bucket-level locking (segment locking), reducing contention vs. a single global lock
 - However, repeated `put()` calls on the same `Ball` key still incur synchronization overhead
- **Read-Write Visibility:**
 - After `invokeAll().get()`, all field writes in tasks are visible to the main `GameLoop` (happens-before guarantee)
 - The Java Memory Model ensures that synchronization actions (lock acquire/release) create memory barriers
 - Without these barriers, updates to `Ball.vel` and `Ball.pos` made by one task might not be visible to the next phase
- **Task Isolation:**
 - `BallUpdateTask` accesses `board` (the `Board` object) to query boundary constraints
 - This creates a shared dependency: if `Board` is modified during task execution, physics updates may read stale boundary data
 - Current implementation assumes `Board` state (bounds, player positions) is immutable during a phase
 - If asynchronous modifications occur (e.g., player ball movement), visibility must be explicitly synchronized
- **Exception Handling and Task Failure:**
 - If a task throws an unchecked exception, `f.get()` will re-throw it via `ExecutionException`
 - The `GameLoop` must handle `InterruptedException` (task thread interrupted) and `ExecutionException` (task computation failed)
 - Without proper handling, the entire game loop may crash
 - Current implementation in `GameLoop.run()` catches both exceptions and breaks the loop gracefully

2 Design and Architecture

2.1 Architectural Overview for Multithreading

The concurrent Pool implementation adopts an **MVC (Model-View-Controller)** architectural pattern with multiple active components:

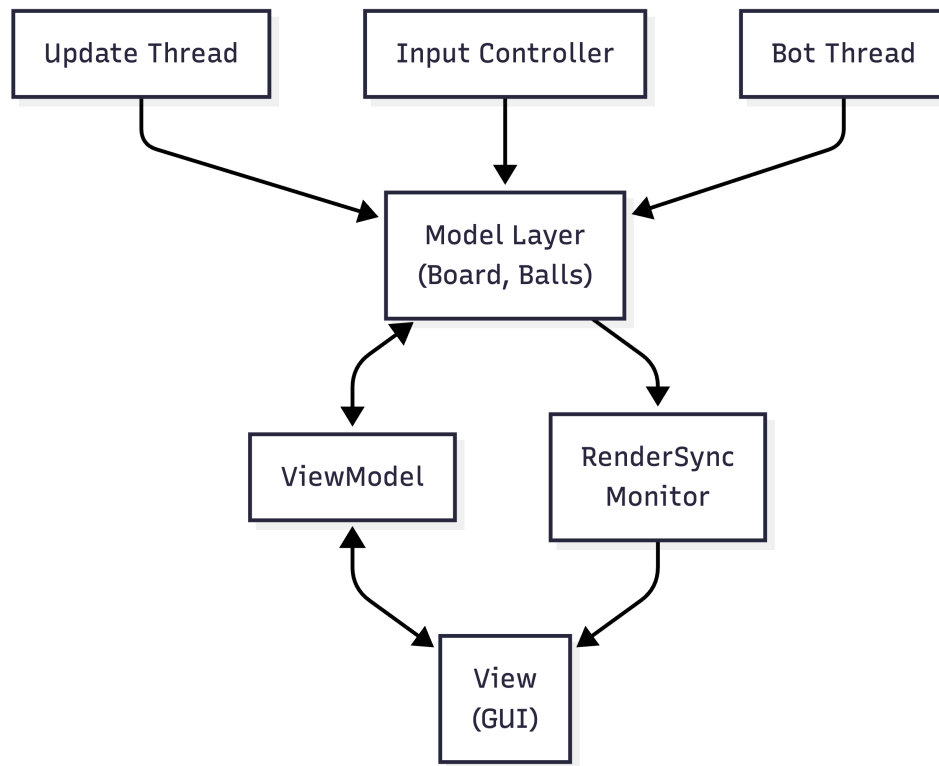


Figure 1: A descriptive caption for your image.

2.1.1 Component Description

Model Layer

The Model layer encapsulates the game state and physics engine:

- **Board:** Central synchronized object maintaining:
 - List of balls with positions and velocities
 - Player ball states (human and bot)
 - Scores for both players
 - Game-over state
 - Ball-to-player touch tracking
- **Ball:** Represents physical entities with mass, radius, position, and velocity
- **Boundary:** Defines board limits for collision detection
- **Monitor (Board):** Synchronized access ensures atomic game state updates

Controller Layer

The Controller comprises two main active components:

- **Update Thread:** Maintains the main game loop
 - Periodically updates board physics (elapsed time-based)

- Synchronizes with View for rendering
- Maintains frame rate statistics
- Checks game-over conditions
- **Bot Thread:** Autonomous player agent
 - Monitors bot ball velocity
 - Generates random kick actions when ball is idle ($v < 0.05$)
 - Waits on bot monitor for notifications
 - Respects game-over state

View Layer

- **ViewFrame:** Swing JFrame managing GUI rendering
- **ViewModel:** Passive data holder with synchronized access
- **RenderSync:** Monitor pattern implementation for synchronous rendering
- **Keyboard Input Handler:** Asynchronously captures user input

2.2 Synchronization Strategy

2.2.1 Monitor-Based Synchronization

The implementation uses Java monitors (synchronized blocks/methods) rather than low-level primitives:

```

1 public synchronized void updateState(long dt) {
2     // Atomic physics computation
3     for (Ball b : balls) {
4         b.updateState(dt, this);
5     }
6     // Atomic collision detection
7     for (int i = 0; i < balls.size(); i++) {
8         for (int j = i + 1; j < balls.size(); j++) {
9             Ball.resolveCollision(balls.get(i), balls.get(j));
10        }
11    }
12 }

```

Listing 2: Board Synchronized Methods

2.2.2 Bot Monitor

A dedicated monitor controls bot behavior synchronization:

```

1 synchronized (board.getBotMonitor()) {
2     while (isRunning()) {
3         var p2 = board.getPlayer2();
4         if (p2.getVel().abs() < 0.05) {

```



```
5         // Generate kick action
6     } else {
7         try {
8             board.getBotMonitor().wait();
9         } catch (InterruptedException e) {
10            // Handle interruption
11        }
12    }
13 }
14 }
```

Listing 3: Bot Thread Synchronization

2.2.3 Rendering Synchronization (RenderSync)

Custom monitor ensures rendering completion before continuing state updates:

```
1 public void render() {
2     long nf = sync.nextFrameToRender();
3     panel.repaint();
4     if (!SwingUtilities.isEventDispatchThread()) {
5         try {
6             sync.waitForFrameRendered(nf);
7         } catch (InterruptedException ex) {
8             ex.printStackTrace();
9         }
10    }
11 }
```

Listing 4: RenderSync Implementation

2.3 Thread Lifecycle

The game execution follows this sequence:

1. **Initialization:** Board configured with balls, boundaries, players
2. **Thread Creation:** Update and Bot threads spawned as daemon threads
3. **Game Execution:**
 - Update thread cycles main game loop
 - Bot thread waits on monitor, activates on velocity threshold
 - Both threads access shared Board state via synchronization
4. **Termination:** Game-over flag set, threads interrupted and stopped

2.4 Architectural Overview for Task-Based Approach

The task-based concurrent implementation differs fundamentally from the thread-per-entity model. Rather than creating persistent threads for each game component, this approach uses the **Executor Framework** with a fixed-size thread pool to parallelize computation-heavy phases of the game loop. The **GameLoop** acts as a master coordinator, delegating parallelizable tasks to a **WorkerPool** abstraction that encapsulates all **ExecutorService** details.

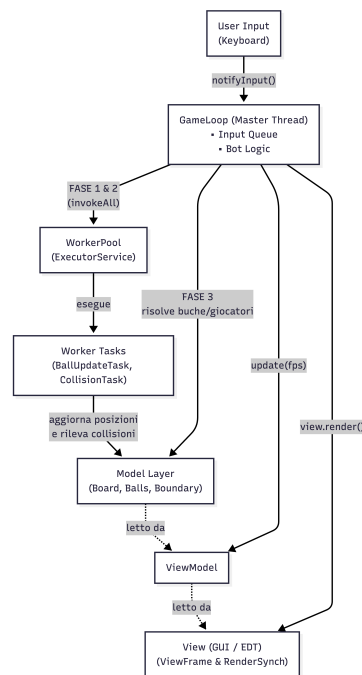


Figure 2: A descriptive caption for your image.

The execution follows a structured four-phase pattern within each game loop iteration:

1. **Phase 1 (Parallel Ball Physics):** `workerPool.parallelBallUpdate()` distributes ball state updates across worker threads
2. **Phase 2 (Parallel Collision Detection):** `workerPool.parallelCollisionDetection()` distributes pairwise collision checks
3. **Phase 3 (Parallel Player Updates):** `workerPool.parallelPlayerUpdates()` distributes player state updates
4. **Phase 4 (Sequential Hole Logic)**

This hybrid approach balances parallelism for expensive operations with simplicity for complex state dependencies.

2.4.1 Component Descriptions

The task-based architecture consists of the following key components:

GameLoop (Master/Orchestrator Thread)

The `GameLoop` extends `Thread` and serves as the main control flow:

- **Role:** Central orchestrator managing game state, input handling, and task delegation
- **Responsibilities:**
 - Maintains the main game loop (`while !board.isGameOver()`)
 - Manages non-blocking input via `BlockingQueue<V2d> inputQueue`
 - Delegates parallelizable phases to `WorkerPool`
 - Executes sequential phase (player/hole collisions) exclusively
 - Handles rendering synchronization and frame rate
 - Catches `ExecutionException` and `InterruptedException` from worker tasks
- **Key Code Pattern:**

```
1 while (!board.isGameOver()) {  
2     V2d impulse = inputQueue.poll(); // Non-blocking input  
3  
4     // Phase 1: Parallel physics  
5     workerPool.parallelBallUpdate(board.getBalls(), elapsed,  
6         board);  
7  
8     // Sequential player logic  
9     if (board.getPlayer1() != null)  
10         board.getPlayer1().updateState(elapsed, board);  
11  
12     // Phase 2: Parallel collision detection  
13     workerPool.parallelCollisionDetection(board.getBalls(),  
14         board.getLastTouchedBy());  
15  
16     // Phase 3: Parallel player updates  
17     workerPool.parallelPlayerCollision();  
18  
19     board.checkHolesAndEndGame();  
20  
21     // Render  
22     viewModel.update(board, fps);  
23     view.render();  
24 }
```

Listing 5: `GameLoop` Main Loop Structure

WorkerPool (Task Coordinator Abstraction)

The `WorkerPool` encapsulates all `ExecutorService` framework details from the `GameLoop`:

- **Role:** Hides complexity of task scheduling, execution, and synchronization

- **Internal Structure:**

- `ExecutorService executor`: Fixed thread pool with size $k = 2 \times \text{availableProcessors}$
- Three public methods: `parallelBallUpdate()`, `parallelCollisionDetection()`, and `parallelPlayerCollision()`
- All use `invokeAll()` for barrier synchronization

- **Key Guarantee:** The `GameLoop` does not import `java.util.concurrent.*`, maintaining clean separation of concerns

- **Exception Propagation:** Exceptions in tasks are bundled into `ExecutionException` and re-thrown at `f.get()`

BallUpdateTask (Parallel Worker)

Implements `Callable<Void>` to update a range of balls' physics state:

- **Input Parameters:**

- `List<Ball> balls`: Shared reference to all balls
- `int from, int to`: Range $[from, to)$ of balls this task processes
- `long dt`: Elapsed time since last frame
- `Board board`: Context for boundary constraints

- **Computation:**

```
1 public Void call() {  
2     for (int i = from; i < to; i++) {  
3         balls.get(i).updateState(dt, board);  
4     }  
5     return null;  
6 }
```

Listing 6: BallUpdateTask Execution

- **Synchronization:** No locks required; each task reads from distinct list indices and modifies only the `Ball` objects at those indices. The `Board`'s boundary data is assumed immutable during execution.

CollisionTask (Parallel Worker)

Implements `Callable<Void>` to resolve pairwise ball collisions:

- **Input Parameters:**

- `List<Ball> balls`: Shared reference to all balls
- `int from, int to`: Range $[from, to)$ of "first" balls in pairs
- `ConcurrentHashMap<Ball, Integer> lastTouchedBy`: Shared map of ball-to-player touches

- **Pair Enumeration:** For each ball $i \in [from, to)$, checks pairs (i, j) where $j \in [i + 1, n)$
- **Synchronization Protocol:**

```

1  for (int i = from; i < to; i++) {
2      for (int j = i + 1; j < n; j++) {
3          Ball a = balls.get(i);
4          Ball b = balls.get(j);
5
6          // Ordered lock acquisition (deadlock prevention)
7          Ball first = System.identityHashCode(a) <= System.
            identityHashCode(b) ? a : b;
8          Ball second = (first == a) ? b : a;
9
10         synchronized (first) {
11             synchronized (second) {
12                 Ball.resolveCollision(a, b);
13                 // Update lastTouchedBy if collision occurred
14                 if (velocities changed) {
15                     lastTouchedBy.put(a, 0);
16                     lastTouchedBy.put(b, 0);
17                 }
18             }
19         }
20     }
21 }

```

Listing 7: Ordered Lock Acquisition in CollisionTask

- **Fine-Grained Locking:** Each pair's collision resolution is protected by locks on both Ball objects, but independent pairs may execute concurrently.

PlayerCollisionTask (Parallel Worker)

Implements `Callable<Void>` to resolve player-ball collisions in parallel:

- **Input Parameters:**
 - `List<Ball> balls`: Shared reference to all balls
 - `int from, int to`: Range $[from, to)$ of "first" balls in pairs
 - `ConcurrentHashMap<Ball, Integer> lastTouchedBy`: Shared map of ball-to-player touches
- **Pair Enumeration:** For each ball $i \in [from, to)$, checks pairs (i, j) where $j \in [i + 1, n)$
- **Synchronization Protocol:**

```

1  for (int i = from; i < to; i++) {
2      Ball b = balls.get(i);
3

```

```

 4      // Collisione con player1
 5      if (player1 != null) {
 6          V2d vPrima = b.getVel();
 7          Ball.resolveCollision(player1, b);
 8          if (!b.getVel().equals(vPrima)) {
 9              lastTouchedBy.put(b, 1);
10          }
11      }
12
13      // Collisione con player2
14      if (player2 != null) {
15          V2d vPrima = b.getVel();
16          Ball.resolveCollision(player2, b);
17          if (!b.getVel().equals(vPrima)) {
18              lastTouchedBy.put(b, 2);
19          }
20      }
21  }
22  return null;

```

Listing 8: Ordered Lock Acquisition in PlayerCollisionTask

Task Decomposition

Task decomposition in the Pool implementation follows a **range-partitioning** strategy:

- **Chunk Size Calculation:**

```

1  int nWorkers = Runtime.getRuntime().availableProcessors() *
    2;
2  int chunkSize = Math.max(1, n / nWorkers);

```

Listing 9: Chunk Size Computation in WorkerPool

where n is the number of balls. This ensures:

- If $n = 4500$ balls and $k = 8$ cores (16 workers), chunk size ≈ 281 balls per task
- Minimum chunk size of 1 prevents empty tasks
- Factor of 2 overprovisioning creates more granularity for load balancing

2.5 Synchronization Strategy

Task-based synchronization employs a layered strategy combining barrier synchronization with fine-grained locking:

2.5.1 Barrier Synchronization via `invokeAll()`

Parallel phases use `ExecutorService.invokeAll()`, which provides implicit barrier semantics:

- **Behavior:** Blocks until *all* submitted tasks complete

- **Java Memory Model Guarantee:** All field writes in tasks are visible after `invokeAll()` returns (happens-before relationship)
- **Game Loop Implications:**
 - Phase 1 completes $\implies$ all ball positions/velocities updated and visible
 - Phase 2 completes $\implies$ all collision resolutions applied and visible
 - Phase 3 completes $\implies$ all player updates applied and visible
 - Sequential Phase 4 executes with consistent global state
- **Performance Cost:** Tail latency — the slowest task determines phase duration

2.5.2 Fine-Grained Ball Locking in Collision Detection

`CollisionTask` uses nested synchronized blocks to prevent data races during collision resolution:

- **Why Two Locks:** Each `Ball.resolveCollision(a, b)` modifies both `Ball` objects' velocity and position. Without synchronization, a concurrent update might corrupt the ball state.
- **Deadlock Prevention via Ordering:**
 - All code acquires locks in a consistent order: always lock Ball A before Ball B if `identityHashCode(A) ≤ identityHashCode(B)`
 - Since identity hash codes are fixed per object, this prevents circular wait (requirement for deadlock)
- **Lock Contention:** Multiple tasks may compete for the same Ball locks
 - If two tasks both need to resolve collisions involving the same Ball, one task waits
 - This serializes independent collisions unnecessarily, but ensures safety
 - In practice, contention is moderate since each task processes a different range of "first" balls

2.5.3 ConcurrentHashMap for Shared State

The `lastTouchedBy` map is a `ConcurrentHashMap<Ball, Integer>` to allow concurrent updates from multiple `CollisionTask` instances:

- **Concurrent Put Operations:** Multiple tasks call `lastTouchedBy.put(ball, 0)` simultaneously
- **Bucket-Level Locking:** `ConcurrentHashMap` uses segment-based locking (in modern Java, striped locking)
 - Reduces contention compared to a single global synchronized lock
 - Different Ball keys typically hash to different buckets, allowing concurrent puts
 - Same-bucket updates still serialize
- **Visibility Guarantee:** All puts are visible globally via `ConcurrentHashMap`'s internal synchronization

2.5.4 Exception Handling and Reliability

Task-based execution requires careful exception handling to prevent loop crashes:

- **ExecutionException:** If a task throws an unchecked exception, `f.get()` re-throws it wrapped in `ExecutionException`
- **InterruptedException:** If the `GameLoop` thread is interrupted (e.g., window closed), waiting on futures throws `InterruptedException`
- **GameLoop Protection:**

```
1 try {  
2     workerPool.parallelBallUpdate(board.getBalls(), elapsed,  
3         board);  
4     workerPool.parallelCollisionDetection(board.getBalls(),  
5         board.getLastTouchedBy());  
6     workerPool.parallelPlayerCollision();  
7 } catch (InterruptedException | ExecutionException e) {  
8     Thread.currentThread().interrupt();  
9     break; // Exit game loop gracefully  
10 }
```

Listing 10: Exception Handling in `GameLoop`

- **Graceful Degradation:** Catching exceptions allows the game loop to terminate cleanly rather than crashing the entire JVM

2.5.5 Memory Visibility and Happens-Before Guarantees

The Java Memory Model ensures correctness through explicit synchronization points:

1. **Task Submission:** When a task is submitted to the executor, all field writes in the `GameLoop` thread are visible to the task
2. **Task Execution:** Workers read shared `Ball` objects and `Board` state; without locks, visibility is not guaranteed
3. **invokeAll() Barrier:** When `invokeAll()` returns, all task writes are visible to the `GameLoop` thread
4. **Nested Locks in Collisions:** Each `synchronized` block creates a memory barrier, ensuring `Ball` field updates are visible across tasks

This ensures that despite the distributed nature of tasks, the game state remains consistent across phase boundaries.

3 Behavior Specification: Petri Nets

3.1 High-Level Game Flow Net

The overall game behavior is modeled using a Petri Net capturing the major state transitions:

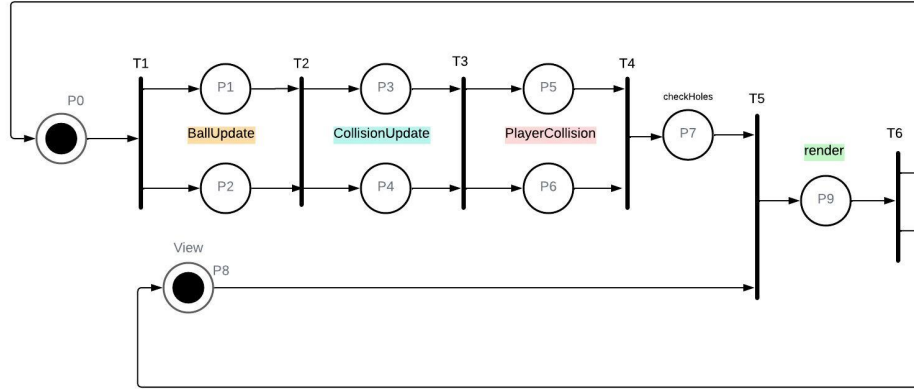


Figure 3: High-Level Game Flow Petri Net

4 Performance Evaluation

4.1 Experimental Setup

4.1.1 Test Configurations

Three board configurations are tested:

Configuration	Small Balls	Expected Complexity
Minimal	2	
Large	400	
Massive	4,500	

Table 1: Board Configurations and Collision Detection Complexity

4.1.2 Performance Metrics

- **Frame Rate (FPS):** Frames rendered per second
- **Update Latency:** Time to compute one physics update cycle
- **CPU Utilization:** Percentage of available cores used
- **Throughput:** Simulation steps per second
- **Speedup:** Concurrent vs. sequential frame rate ratio

4.2 Expected Performance Improvements

4.2.1 Sequential Baseline Issues

The sequential implementation exhibits:

- Minimal/Large: fps (acceptable)
- Massive: 2 fps (unacceptable frame rate degradation)

4.2.2 Concurrent Improvements

Update Thread Parallelization:

- Potential to parallelize ball state updates (independent operations)
- Potential to parallelize collision detection
- Rendering no longer blocks physics computation

Bot Thread Independence:

- Bot decisions computed asynchronously
- Reduces decision latency from main game loop
- Enables responsive user input even during intensive physics

Expected Speedup Formula:

$$S = \frac{T_{seq}}{T_{conc}} = \frac{f_{seq}}{f_{conc}}$$

For massive configuration with parallel collision detection over k cores:

$$S \approx 1 + \frac{(O(n^2) - O(n))}{k}$$

With $n = 4,500$ balls and $k = 4$ cores:

$$S \approx 1 + \frac{10.1M - 4,500}{4} \approx 2.5\text{-}2.8\times$$

// Note: Actual speedup may be lower due to synchronization overhead and non-parallelizable code sections.

4.2.3 Task-based Improvements

5 Formal Verification with JPF

5.1 Java Path Finder Integration

JPF (Java Path Finder) is a model checker developed at NASA for verifying Java programs. The Pool implementation integrates JPF verification through configuration and test drivers.

5.1.1 JPF Configuration

Key aspects of JPF application to Pool:

```

1 target=pcd.assignment01.controller.MainJPF
2 listener = gov.nasa.jpf.listener.PropertyListener

```

Listing 11: JPF Verification Approach

5.2 Safety Properties Verified

5.2.1 1. Mutual Exclusion (No Data Races)

Property: Board state modifications are atomic and do not interleave.

Verification Method:

- JPF tracks synchronized block boundaries
- No two threads can execute synchronized Board methods concurrently
- All field accesses to mutable state (balls, scores) occur within synchronization

Critical Section Analysis:

```

1 public synchronized void updateState(long dt) {
2     // CRITICAL SECTION START
3     for (Ball b : balls) { /* ... */ } // Protected access to
        balls
4     score1++; // Protected access to score
5     // CRITICAL SECTION END
6 }

```

Listing 12: Critical Section: Board.updateState()

5.2.2 2. Deadlock Freedom

Property: No circular wait-for cycles among threads.

Lock Hierarchy:

Lock Order	Acquisition Pattern
1. Board	Always acquired first (main game state)
2. Bot Monitor	Acquired only from within Bot thread
3. RenderSync	Acquired from Update thread only

Table 2: Lock Hierarchy for Deadlock Prevention

Analysis:

- Bot thread never acquires Board after waiting on botMonitor
- No wait chains among Update, Bot, and EDT threads
- Circular dependencies are structurally impossible

5.2.3 3. Liveness (Game Termination)

Property: Game always terminates within bounded time steps.

Termination Conditions:

1. **Score-based:** All small balls scored or removed
2. **Player-based:** A player's ball enters a hole

3. User-initiated: Window closed or game stopped

JPF Verification:

```

1 public void testGameTermination() {
2     // JPF will explore all possible execution paths
3     // and verify that all paths eventually reach:
4     assert !isRunning() : "Game must terminate";
5     assert gameOverFlag : "Game-over state must be set";
6 }

```

Listing 13: Termination Assertion

5.3 Race Condition Detection

5.3.1 Synchronized Field Accesses

JPF verifies that all shared mutable fields are protected:

```

1 private List<Ball> balls;           // Protected by synchronized
   methods
2 private int score1;                 // Protected by synchronized
   methods
3 private boolean running;           // Protected by synchronized
   getter/setter
4 private boolean isGameOver;         // Protected by synchronized
   methods

```

Listing 14: Synchronized Field Protection

5.3.2 Temporal Assertions

Critical invariants verified during execution:

```

1 // Invariant 1: Score consistency
2 assert score1 >= 0 && score2 >= 0 : "Scores are non-negative";
3
4 // Invariant 2: Ball list validity
5 assert balls != null : "Ball list always exists";
6 assert balls.stream().allMatch(b -> b != null) :
7     "No null balls in list";
8
9 // Invariant 3: Game state consistency
10 assert !(isGameOver && !running) :
11     "Game-over implies not running";

```

Listing 15: Invariants Checked by JPF

5.4 Model Checking Results

5.4.1 Test Configuration: MainJPF

The MainJPF class provides a minimal test case for JPF verification:

```
1 public class MainJPF {
2     public static void main(String[] args) {
3         // Minimal board configuration for JPF
4         Board board = new Board();
5         board.init(new MinimalBoardConf());
6
7         GameController controller =
8             new GameController(board, null, null);
9
10        // JPF explores all possible interleavings
11        controller.startGame();
12
13        // Verification terminates after bounded steps
14        try {
15            Thread.sleep(100);
16        } catch (InterruptedException e) {
17            // Expected: JPF may interrupt
18        }
19
20        controller.stopGame();
21    }
22 }
```

Listing 16: MainJPF Verification Driver

References

- [1] NASA Formal Methods Program, *Java Path Finder - Model Checker for Java Programs*, <https://javapathfinder.github.io/>
- [2] J.L. Peterson, *Petri Nets*, *ACM Computing Surveys*, vol. 9, no. 3, 1977.
- [3] C.A.R. Hoare, *Monitors: An Operating System Structuring Concept*, *Communications of the ACM*, 1974.
- [4] B. Goetz et al., *Java Concurrency in Practice*, Addison-Wesley, 2006.